

 E-COMMERCE
RETURN RATE

 OVERVIEW

 SEGMENTS

 RISK SCORE
ANALYSIS

 INSIGHTS

 RESET FILTERS

E-COMMERCE RETURN RATE ANALYSIS DASHBOARD

Product Category

All

Return Status

All

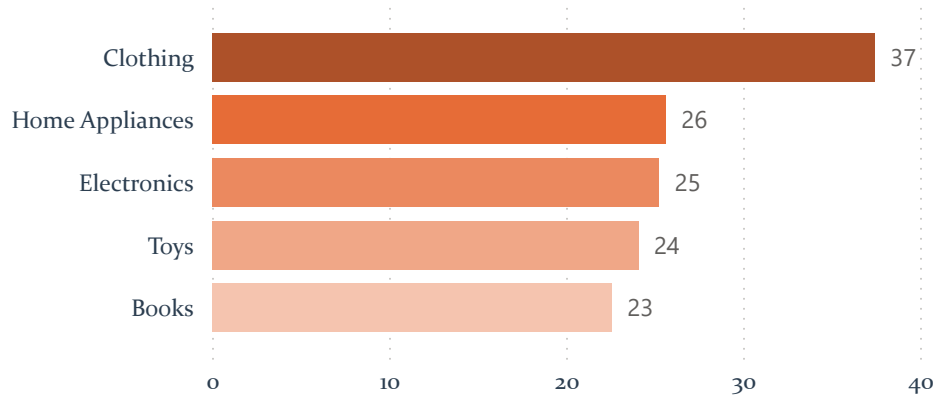
5K
Total Orders

29.00
Return Rate %

290K
Total Return Cost

11.54M
Total Profit

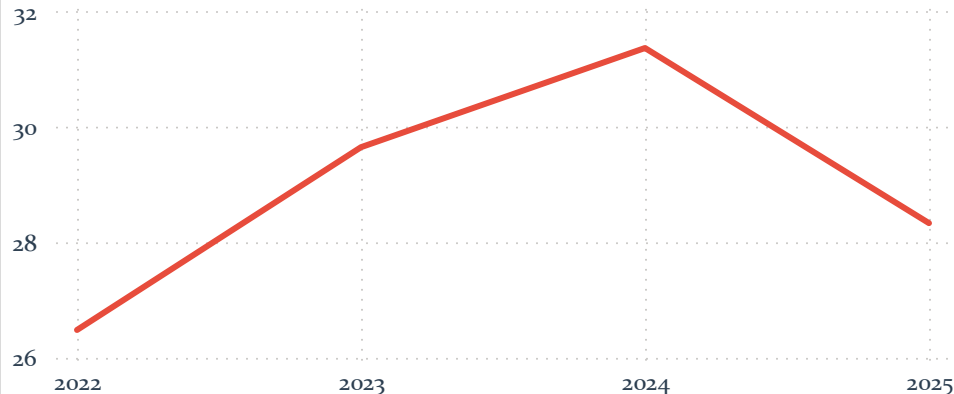
Return Rate % by Product Category



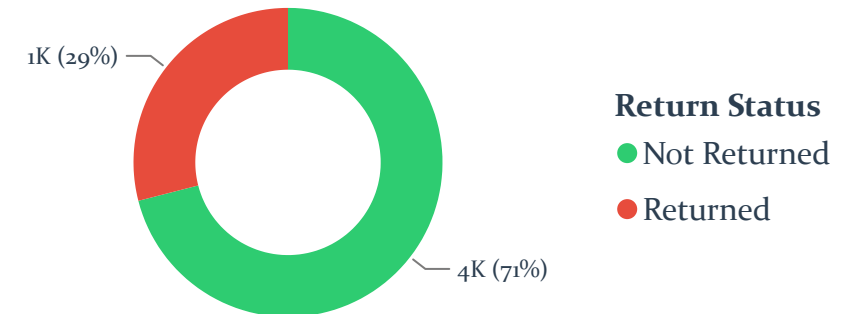
Snapshot Insights (Overview)

- Overall return rate is 29% — nearly 1 in 3 orders returned.
- Clothing has the highest return rate at (37.4%).
- Books category has the lowest return rate (22.6%).
- Return rate peaked in 2024 (~31%), followed by a decline in 2025.
- 71% of all orders delivered without returns.

Return Rate Trend by Year



Return Status Distribution



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Shipping Method

All

Payment Method

All

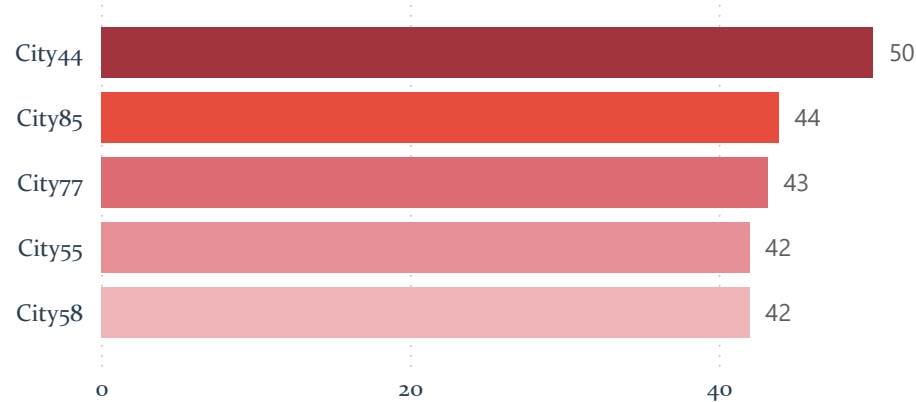
0.50
Avg Risk Score

290K
Total Return Cost

2K
High Risk Orders

1K
Total Returned

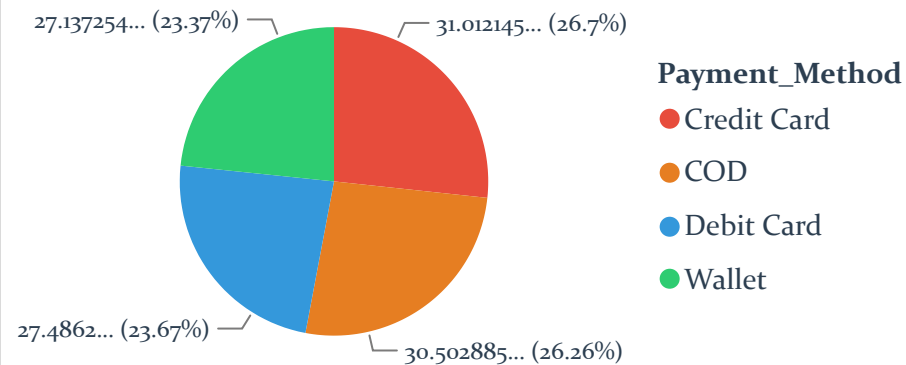
Top 5 High Risk Cities



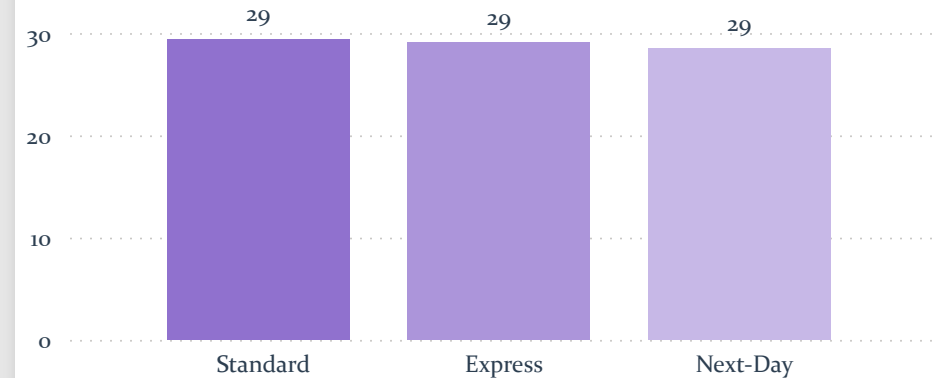
KEY GEOGRAPHIC & CHANNEL INSIGHTS

- City44 has the highest return rate (50%), followed by City85 (43.9%) and City77 (43.2%)
- Credit Card has the highest return rate (31%), with COD close behind (30.5%)
- Standard shipping shows a slightly higher return rate than other methods
- Returns are concentrated in a few cities, indicating regional issues

Return Rate by Payment Method



Return Rate by Shipping Method



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Risk Label

All

Product Category

All

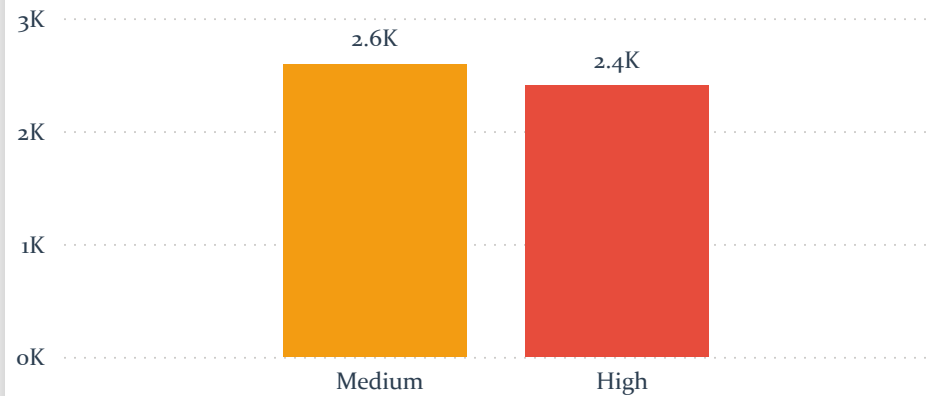
0.50
Avg Risk Score

2K
High Risk Orders

575
Very High Risk Orders

1K
Total Returned

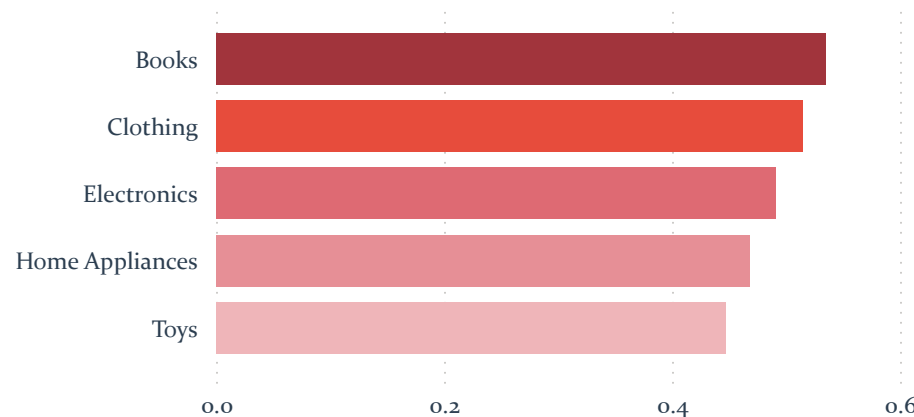
Orders by Risk Label



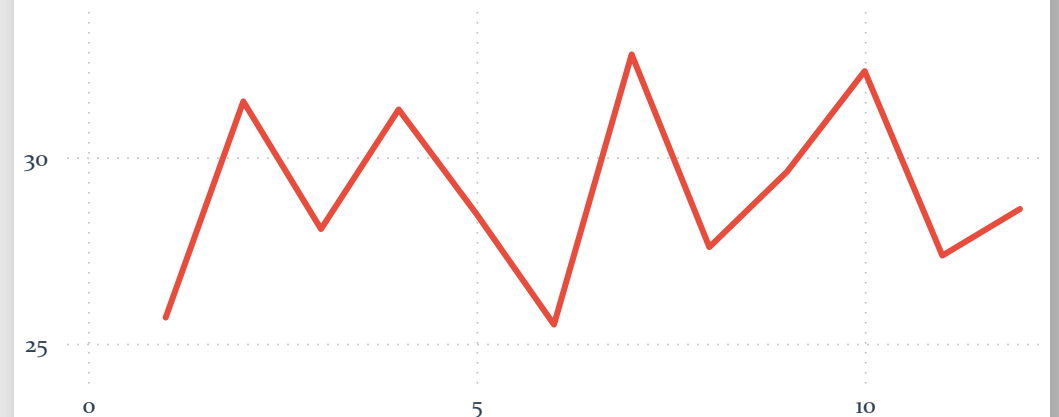
RISK SCORE INSIGHTS

- 2,406 orders are classified as High Risk
- Clothing products have the highest average risk score
- Credit Card orders show the highest average risk score
- The ML model achieves 71% accuracy on test data
- Risk scoring helps identify high-risk orders before dispatch
- Return rates peaked in mid-2024, indicating possible seasonal trends

Avg Risk Score by Category



Return Rate Trend by Month



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Risk Label

All

Product Category

All

2K
High Risk Orders

575
Very High Risk Orders

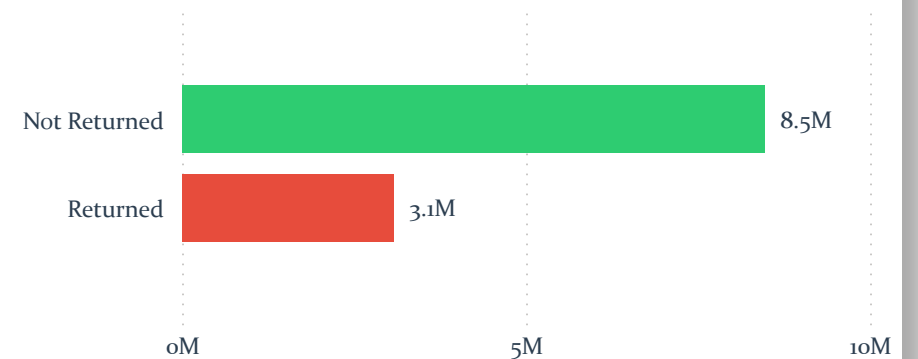
290K
Total Return Cost

11.54M
Total Profit

High Risk Orders Detail

Product_Category	User_Location	Payment_Method	Return_Risk_Score	Risk_Label
Clothing	City20	Wallet	0.49	Medium
Clothing	City24	Credit Card	0.49	Medium
Clothing	City4	Wallet	0.49	Medium
Clothing	City72	Wallet	0.49	Medium
Clothing	City73	COD	0.49	Medium
Electronics	City33	Credit Card	0.49	Medium
Electronics	City66	Credit Card	0.49	Medium
Home Appliances	City83	Credit Card	0.49	Medium

Profit vs Return Status



BUSINESS RECOMMENDATIONS

- Clothing needs urgent quality control — 37.4% return rate
- Investigate City44 — 50% return rate detected
- Review suppliers with high defective product returns
- Add extra verification for Credit Card and COD orders
- Reduce heavy discounts on Clothing category
- Use ML risk scores to flag high risk orders before dispatch
- Focus retention efforts on High and Very High risk orders

Total Return Cost by Category

